

화학수학

스물한 번째 수업

시험 안내 및 복습

기말고사

- 시간: 12월 16일(목) 오후 1시 반~2시 반
- 장소: 화학관 315호(중간고사 시험장소)
- 계산기 및 신분증 지참
- 10분 전까지 입실 완료
- 시험 범위: 미분방정식 / 연산자, 행렬, 군론 / 확률 및 통계
- 수업 시간에 다루지 않은 내용은 범위에 포함되지 않음

12. Differential Equations

(12장 미분방정식)

Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
 - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$

Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions

(일반해와 특수해)

- **Solution:** any function that reduces the DE to an identity when substituted into the DE

For a DE of order n ,

- **General solution:** contains n arbitrary constants
- **Particular solution:** contains no arbitrary constants
 - Particular solutions can be generated from the general solution



Additional Conditions

- For a DE of order n , if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first $(n - 1)$ derivatives at a single point
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Solving a Differential Equation

1. Obtain the **general solution**: hardest part!
 2. If initial/boundary conditions are given, substitute the conditions to **determine the coefficients**
- No universal method to solve all DEs
 - For specific forms of DEs, we have general solving methods

First-Order ODEs

$$F\left(y, \frac{dy}{dx}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y) \frac{dy}{dx} = f(x)$$

This DE can be manipulated into

$$g(y) dy = f(x) dx$$

$$\int g(y) dy = \int f(x) dx + C$$

This equation provides the **general solution**



Exact Differential Equations

- Pfaffian form:

$$M(x, y) \, dx + N(x, y) \, dy = 0$$

If the differential equation is **exact**,
there is a function $f(x, y)$ such that

$$df = M(x, y) \, dx + N(x, y) \, dy = 0$$

which implies that $f(x, y) = k$ (constant)

How to Find $f(x, y)$

$$df = M(x, y) dx + N(x, y) dy = 0$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_C df$$

where C is a curve connecting (x_0, y_0) and (x_1, y_1) . Take a rectangular path from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) dx + \int_{y_0}^{y_1} N(x_1, y) dy$$

This gives the relationship between x and y .



Integrating Factors

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **inexact**, but we know the **integration factor** $g(x, y)$ that makes

$$g(x, y)M(x, y) dx + g(x, y)N(x, y) dy = 0$$

an exact differential equation, we can solve this DE.

Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{dy}{dx} + P(x)y = f(x)$$

- General solution:

$$y = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx + C e^{-\int P(x)dx}$$

Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
 - Homogeneous (§ 12.2)
 - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs



Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \dots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \dots, c_k$.

Reduction of Order

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution y_1 is known,
we can obtain the other solution y_2

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Homogeneous Case

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시행해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Three possible cases!

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation →
complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_c + y_p$

Guess for Particular Solution

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$Ae^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t}(A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$

Glitch in the Method

A trial solution y_p can be
a solution of the complementary equation.

In this case, try $x^n y_p$ instead!

(n : smallest integer that resolves the problem)

Laplace Transforms

- Laplace transform

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \mathcal{L}\{f(t)\}$$

- Inverse Laplace transform

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds = \mathcal{L}^{-1}\{F(s)\}$$

requires complex analysis;
We will not use this formula

Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\frac{1}{s-a}$	$e^{at} (s > a)$
$\frac{s}{s^2 + a^2}$	$\cos(at)$
$\frac{a}{s^2 + a^2}$	$\sin(at)$
$\frac{s}{s^2 - a^2}$	$\cosh(at)$
$\frac{a}{s^2 - a^2}$	$\sinh(at)$



Useful Theorems

- Shifting Theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

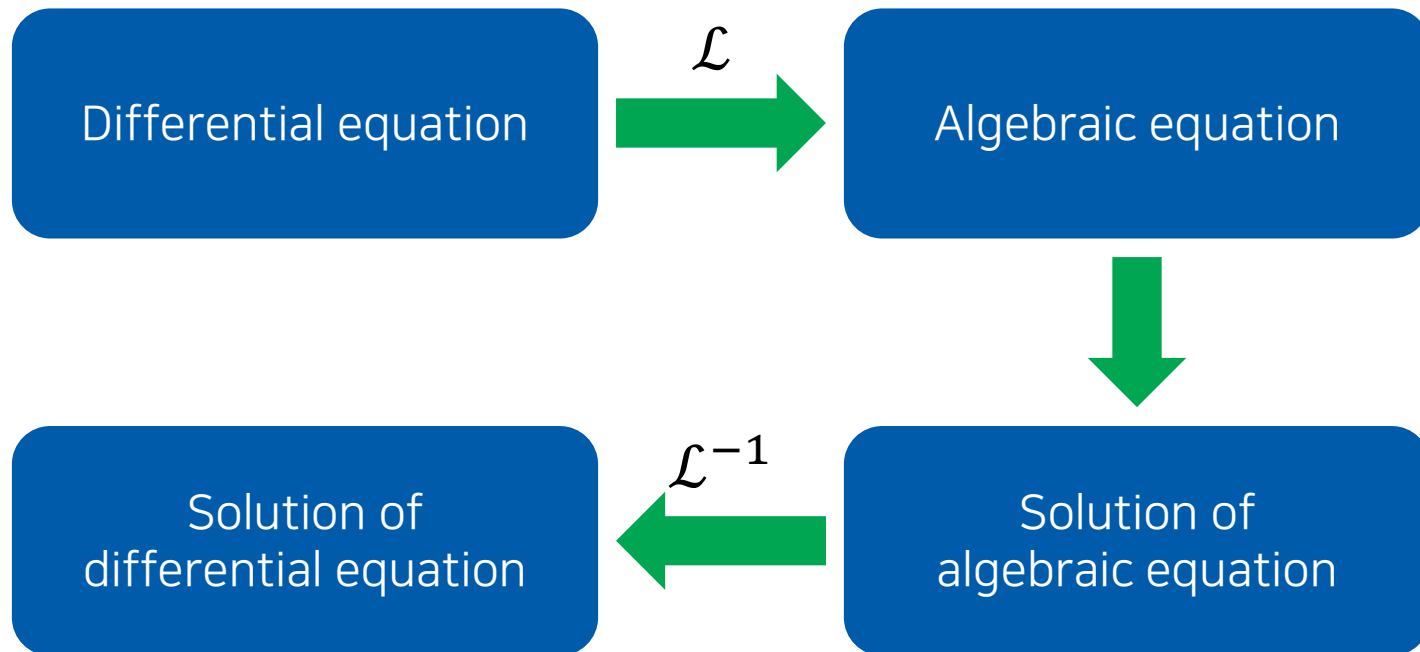
- Derivative theorem

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

- Integral theorem

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}$$

Solving DE with Laplace Transforms



Partial Differential Equations

- General form of a **linear second-order PDE**

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \dots, G$ are functions of x and y .

- $G(x, y) = 0$: **homogeneous**
- $G(x, y) \neq 0$: **inhomogeneous**

Separation of Variables

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

Assume that $u(x, y) = X(x)Y(y)$.

$$\begin{aligned} X' &= dX/dx, X'' = d^2X/dx^2, \dots \\ Y' &= dY/dy, Y'' = d^2Y/dy^2, \dots \end{aligned}$$

If we obtain $f(x, X, X', \dots) = g(y, Y, Y', \dots)$, let

$$f(x, X, X', \dots) = g(y, Y, Y', \dots) = \lambda$$

where λ is the **separation constant**.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

13. Operators, Matrices, and Group Theory

(13장 연산자, 행렬, 군론)

Mathematical Operators (연산자)

perform **mathematical operations** on some function

e.g. "+3" operator, "×2" operator, "d/dx" operator

- Typically denoted by a letter with a **caret** (^)
- Operate on the function on its right

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A} ?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to **commute**
- **Commutator** (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A} \cdots \hat{A} \text{ (} n \text{ times)}$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$
- Not all operators have inverses (*e.g.* $\hat{0}$)

Operator Algebra and Differential Equation

- Derivative operator $\hat{D}_x = d/dx$
- Homogeneous linear DE with constant coefficients

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

$$(a_n \hat{D}_x^n + a_{n-1} \hat{D}_x^{n-1} + \cdots + a_1 \hat{D}_x + a_0 \hat{E})y = 0$$

This can be solved by operator algebra

Eigenfunctions and Eigenvalues

(고유함수와 고윳값)

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an **eigenfunction** of $\hat{A}$, and a is called an **eigenvalue**. This equation is called an **eigenvalue equation** (고윳값 방정식).

Operators in Quantum Mechanics

“For every **observable quantity**,
there is a corresponding **mathematical operator**”

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z)$$

Hermitian Property (에르미트 성질)

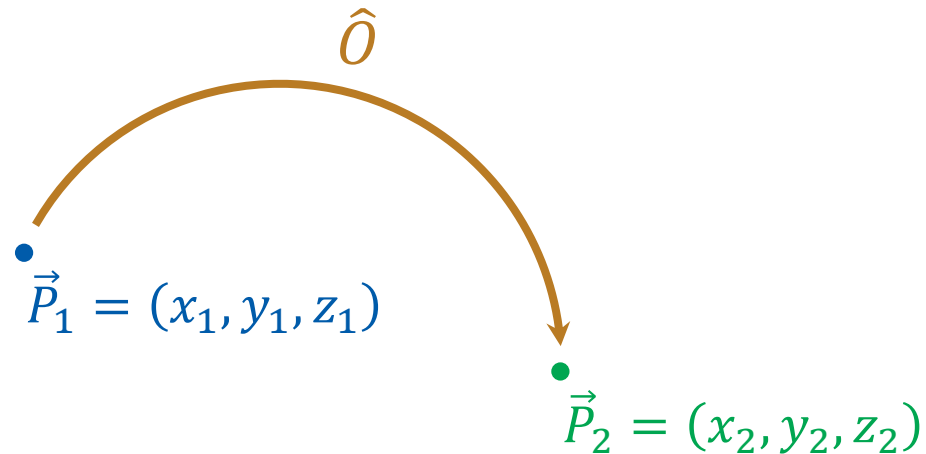
- If $\hat{A}$ is **Hermitian**, it obeys the relation

$$\int \chi^* (\hat{A}\psi) dq = \int (\hat{A}\chi)^* \psi dq$$

where χ and ψ are arbitrary functions and q represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators

(점대칭 연산자)

a class of symmetry operators
that leave **at least one point unmoved**
(the unmoved point = origin)

Identity operator $\hat{E}$

Inversion operator $\hat{i}$

Reflection operator $\hat{\sigma}$

Rotation operator $\hat{C}_n, \hat{S}_n$



Symmetry Elements (대칭 요소)

Point, line, or plane relative to which the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol
- Inversion i : the origin
- Reflection σ : the reflection plane
- Rotation C_n, S_n : the rotation axis

Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, **the symmetry operator belongs to the object.**
- Coordinate system
 - origin: center of mass of the object
 - z axis: the rotation axis of highest order (largest n)

Symmetry Operators on Functions

If an operator $\hat{O}$ operates as $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$,
for a function f ,

$$\hat{O}f = g$$

where $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$.

Use the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) !

Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

the operator leaves a function unchanged

- Eigenvalue = -1

the operator changes the sign of the function's value at all points

Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

matrix elements
(행렬 원소)

(m by n matrix)

Matrix Algebra

- Two matrices are **equal** if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix

- **Sum of two matrices:** $\mathbf{C} = \mathbf{A} + \mathbf{B}$ if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- **Product of a scalar c and a matrix $\mathbf{A}$:** $\mathbf{B} = c\mathbf{A}$ iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

Product of Two Matrices

- Matrix product $\mathbf{C} = \mathbf{AB}$ iff

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

where n is the number of columns in $\mathbf{A}$ and the number of rows in $\mathbf{B}$.

$$(m \text{ by } n \text{ matrix}) \times (n \text{ by } p \text{ matrix}) = (m \text{ by } p \text{ matrix})$$

Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$$\mathbf{AB} \neq \mathbf{BA} \text{ (in some cases)}$$

Full Bag of New Terms

1. **Special matrices:** identity matrix, zero matrix
2. **Partner of a matrix:** inverse, transpose, Hermitian conjugate
3. **Properties of a matrix:** singular, symmetric, Hermitian, orthogonal, unitary
4. **Derived quantities from a matrix:** trace, determinant

Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^n a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

Gauss-Jordan Elimination

- Construct an **augmented matrix** (첨가행렬)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[\begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array} \right]$$

- Then make the identity matrix

$$\left[\begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array} \right] A^{-1}$$

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Applications of the Determinant

- Slater determinant
- Vector product of two vectors
- Jacobian calculation
- Determinant of a singular matrix
- Matrix eigenvalue problem

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

Group (군)

Collection of members
with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the **product** AB is also a member of the group
2. The group must contain the **identity** E such that

$$AE = EA = A \text{ (for every member } A\text{)}$$

3. The **inverse** of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

4. The **associative law** holds: $A(BC) = (AB)C$



Point Group (점군)

Formed by a set of point symmetry operators
belonging to a symmetrical object

Representation of a Point Group

Symmetry operator $\leftrightarrow$ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

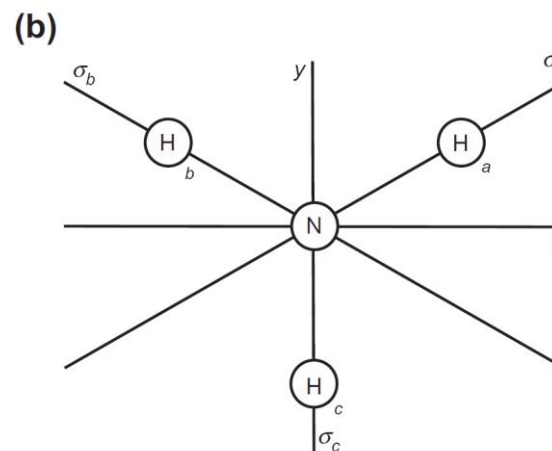
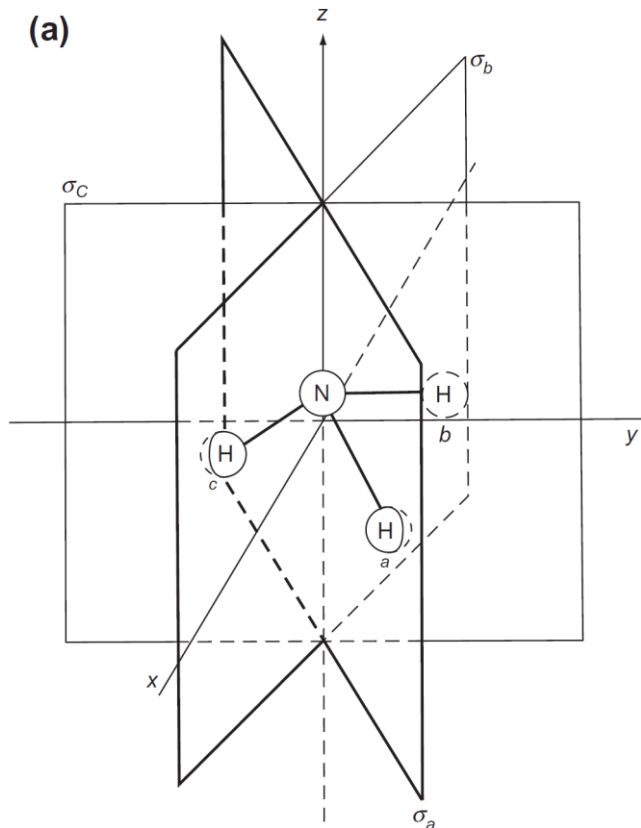
How to Find the Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}y_1 + a_{13}z_1 = x_2 \\ a_{21}x_1 + a_{22}y_1 + a_{23}z_1 = y_2 \\ a_{31}x_1 + a_{32}y_1 + a_{33}z_1 = z_2 \end{cases}$$

From the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) ,
determine a_{ij}

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

Similarity Transformation (답음 변환)

- Two different representations **P** and **Q** in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

when **X** is some square matrix

- Since $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$,

$$\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \text{tr}(\mathbf{XX}^{-1}\mathbf{Q}) = \text{tr}(\mathbf{Q})$$

Character (자표)

- **Character** of a symmetry operator = **trace** of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Character of } \hat{C}_3 = \text{tr } \mathbf{A} = (-1/2) + (-1/2) + 1 = 0$$

Reducible Representation (가약 표현)

All matrices are **block-diagonal**, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{F}$$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal
by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as **irrep**

Character Table (지표표)

- Using group theory theorems, we can find all possible irreps and collect their characters → **character table**
- e.g.* C_{3v} group

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

conventional notation for irreps

15. Probability, Statistics, and Experimental Errors

(15장 확률, 통계, 실험오차)

Probability Theory

study of random events

- **Classical systems:** unknown information
- **Quantum systems:** intrinsic randomness
- **Population:** large set of objects or numbers
 - Classical systems: imaginary set of infinitely many repetitions of a given measurement
 - Quantum systems: imaginary set of infinitely many replicas of the physical system

Probability Distribution (확률분포)

mathematical function that gives
the probabilities of occurrence of
different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포):
possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포):
possible outcomes have continuous values

Probability Distributions

	Discrete	Continuous
Variable	$x_1, x_2, \dots, x_n$	x : real in $[x_{\min}, x_{\max}]$
Probability distribution	p_i	$f(x) dx$
Normalization	$\sum_{i=1}^n p_i = 1$	$\int_{x_{\min}}^{x_{\max}} f(x) dx = 1$
Mean value (normalized dist.)	$\langle x \rangle = \sum_{i=1}^n x_i p_i$	$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x) dx$
Expected value (normalized dist.)	$\langle g \rangle = \sum_{i=1}^n g(x_i) p_i$	$\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x) dx$

Examples and Applications

- Examples

- Binomial probability distribution: discrete
- Uniform probability distribution: continuous
- Gaussian probability distribution: continuous

- Applications

- Quantum mechanics
- Kinetic theory of a gas

Binomial Probability Distribution

- p = probability that “heads” will occur
- $(1 - p)$ = probability that “tails” will occur

$$P(n, m) = \frac{n!}{m! (n - m)!} p^m (1 - p)^{n-m}$$

Mean Value and Variance

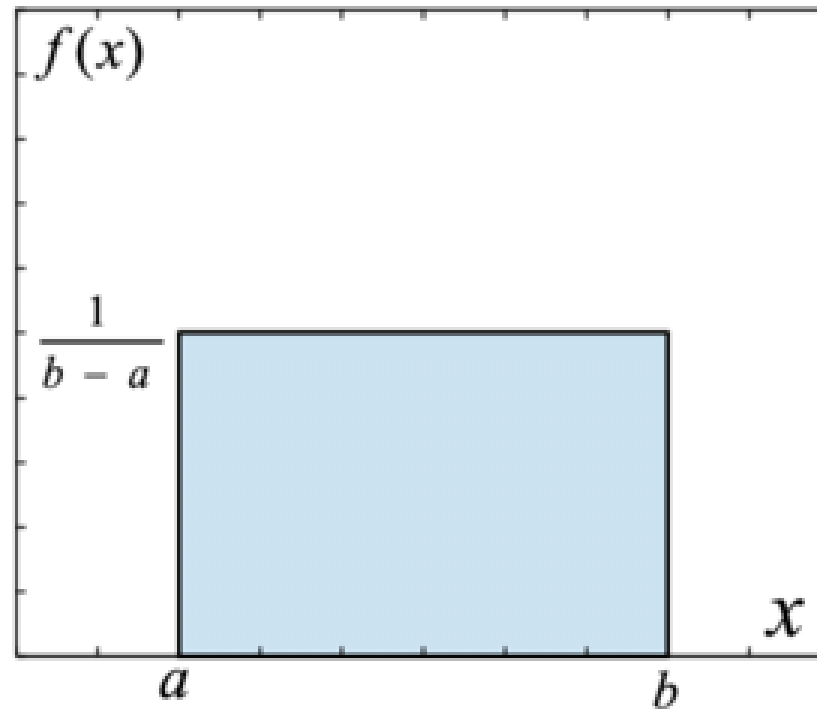
- Mean value

$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i = \sum_{m=0}^n \frac{n!}{m! (n-m)!} p^m (1-p)^{n-m} \times m$$
$$\langle x \rangle = np$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = np(1-p)$$

Uniform Probability Distribution



Mean Value and Variance

- Mean value

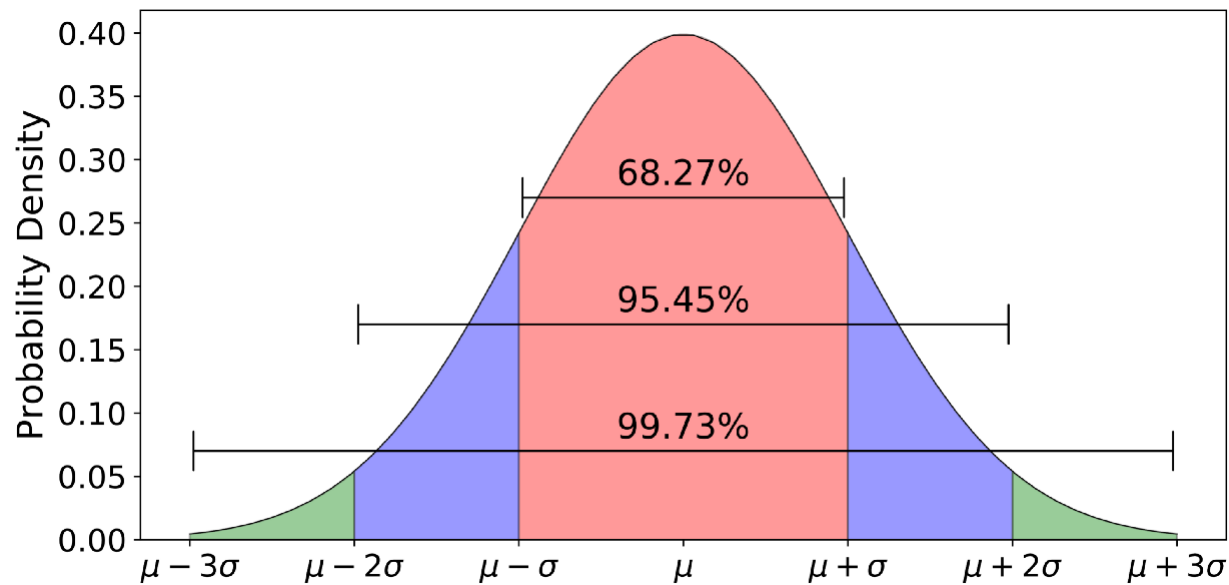
$$\langle x \rangle = \frac{a + b}{2}$$

- Variance

$$\sigma_x^2 = \frac{(a - b)^2}{12}$$

Gaussian Probability Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right]$$



(Image: <https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2>)

Mean Value and Variance

- Mean value

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x) dx = \mu$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \sigma^2$$

Error Function (오차함수)

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

(Fraction between $\mu - \alpha$ and $\mu + \alpha$)

$$\begin{aligned} &= \int_{\mu-\alpha}^{\mu+\alpha} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} dx = \int_{-\alpha/\sqrt{2\sigma^2}}^{\alpha/\sqrt{2\sigma^2}} \frac{1}{\sqrt{\pi}} e^{-t^2} dt \\ &= \frac{2}{\sqrt{\pi}} \int_0^{\alpha/\sqrt{2\sigma^2}} e^{-t^2} dt = \operatorname{erf}\left(\frac{\alpha}{\sqrt{2\sigma^2}}\right) \end{aligned}$$

Probability in Quantum Mechanics

For a wave function ψ of the system,

$$\text{Probability distribution} \propto |\psi|^2$$

e.g. Ground state of a 1-dimensional particle in a box

- Wave function $\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$
- Probability distribution $f(x) = |\psi|^2 = \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right)$

Expectation Value (기댓값)

“The predicted mean value of many measurements of an observable”

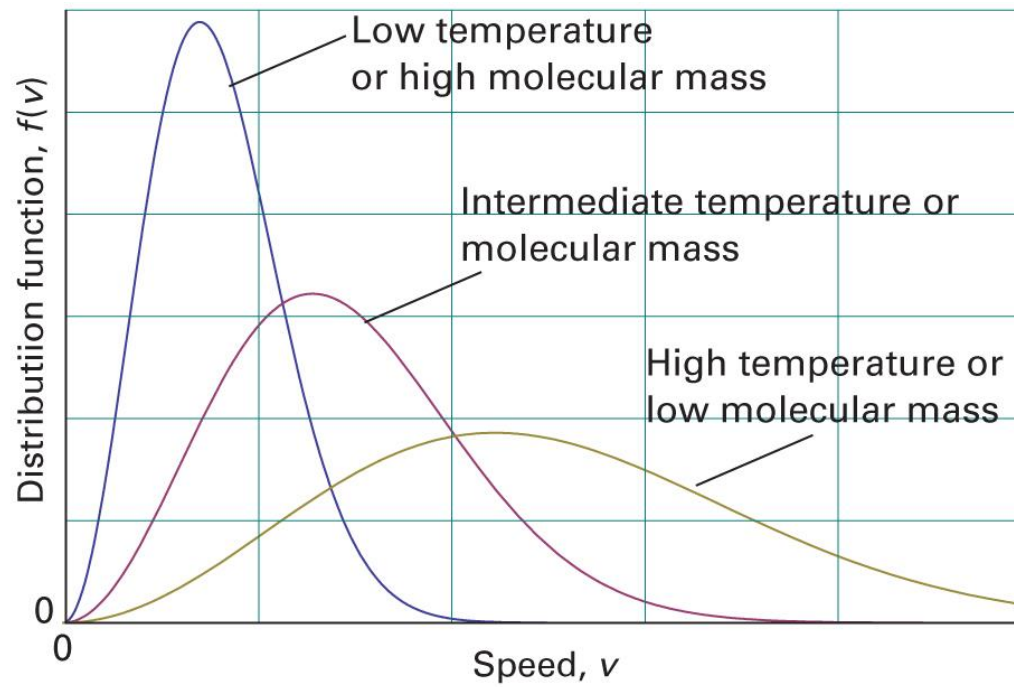
For a wave function ψ of the system,

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, dq$$

where $\hat{A}$ is the operator corresponding to the variable A .

Maxwell-Boltzmann Distribution

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Characteristic Speeds

$$\langle v \rangle = \left(\frac{8k_B T}{\pi m} \right)^{1/2}$$

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \left(\frac{3k_B T}{m} \right)^{1/2}$$

$$v_{\text{mp}} = \left(\frac{2k_B T}{m} \right)^{1/2}$$

Statistics of a Sample

- Goal: estimate mean, variance, standard deviation of the population
- After an infinite number of samplings,
 - $\langle \text{sample mean} \rangle \rightarrow \text{population mean}$
 - $\langle \text{sample variance} \rangle \rightarrow \text{population variance}$
 - $\langle \text{sample standard deviation} \rangle$
 $\rightarrow \text{population standard deviation}$

(unbiased estimates)

Statistics of a Sample

For a sample of N measurements,

- Sample mean: $\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i$
- Sample variance: $s_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \langle x \rangle)^2$
- Sample standard deviation: $s_x = \sqrt{s_x^2}$

Confidence Level (신뢰수준)

- 95% confidence level = errors have a 95% probability of being correct
- If we want to assert that the interval

$$\langle x \rangle - \varepsilon < x < \langle x \rangle + \varepsilon$$

has a 95% chance of containing the true value x_{true} , the interval is called the **95% confidence interval** (95% 신뢰 구간) and ε the **estimated error** (추정 오차)

How to Determine ε

Under some mathematical assumptions, we can derive an estimate for ε :

$$\varepsilon = t(N - 1, \alpha) \frac{s_x}{\sqrt{N}}$$

- N : sample size
- α : significance ($1 - \alpha$ = confidence level)
- $t(N - 1, \alpha)$: Student t value